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ENGR 095

Materials Reflection

In this module, we learned about sensors and the materials and techniques used to make them. We started out with an introduction to the roles of electrical, computer, and materials engineers. We then discussed how each of these engineers would be involved in producing and utilizing sensors. With that background information, we were able to make some guesses as to how we would create our own devices to sense water vapor. I attempted to use two lines of graphite spaced very closely on a paper to detect water vapor, but the resistance of the graphite and the water permeability of the paper made this solution nonfunctional. However, the core idea behind this, using a conductive material laid onto an insulating substrate to make a sensor, has merit. We learned about different techniques to deposit conducting material on top of the substrate, including photolithography and screenprinting. We also learned about the electrical properties of different materials, and how their structure can affect their electrical resistance.

In order to create sensors that function, we toured some of Lehigh's engineering facilities. In Wilbur Powerhouse, we learned how to use a cutting machine to cut out stencils that we could use to make interdigitated electrodes. In the cleanroom, we learned about how spincoating can be used to deposit a thin film of material onto a surface. Using all of this knowledge, we set about actually making our sensors. My group used the stencils we cut out to deposit silver onto glass slides using the screen printing method. We made 3 IDEs, one with silver paint, one with silver paste, and one with copper tape. Then, we used spin coating to cover the new interdigitated electrodes with PEDOT PSS, a material that changes resistivity in response to water vapor. We

used a multimeter to test these sensors. We calculated the resistivity of the silver paint, and observed that the resistance increased when water vapor was present.

Throughout this module, I learned a lot about how material engineers approach problems and use a set of constraints and criteria to select the right materials and processes. It also helped shed light on an area of engineering that I don't often think about. I've used light sensors, resistors, capacitors, and other electrical components in projects before, but I've never given much thought into what causes those things to have the properties that they do. What material in the LDR causes it to respond to light? How do resistors have a stable amount of electrical resistance across a range of temperatures? I'm glad I was able to experience what it's like to actually think about what quality a device needs and then find a way to give it that quality.